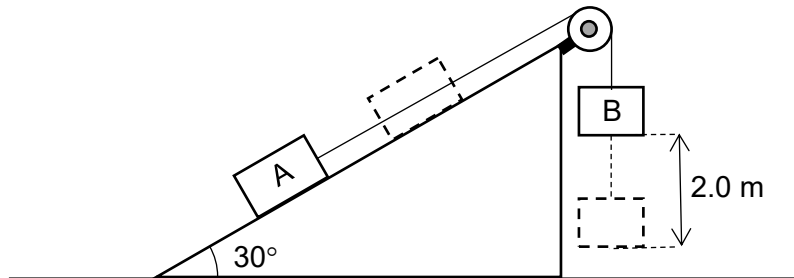


- 7 Blocks A and B, of masses 4.0 kg and 6.0 kg respectively, are connected by a light cord passing over a light, frictionless pulley. Block A is held at rest on a rough slope inclined at 30° to the horizontal as shown.



When A and B are released, block A experiences a constant frictional force of 3.0 N.

What is the total kinetic energy of A and B when B has travelled 2.0 m downwards?

A 39 J

B 72 J

C 78 J

D 118 J

Ans: B

Loss in GPE of B = Gain in GPE of A + Gain in KE of A and B + Work done agst friction

$$6.0(9.81)(2.0) = 4.0(9.81)(2.0 \sin 30) + KE_{A+B} + 3.0(2.0)$$

$$KE_{A+B} = 72 \text{ J}$$

- 8 A wire is stretched elastically by a force of 200 N, causing an extension of 2.00 mm. The force